

SHAMIR KHAN

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[personal-website](#)

Education

Bachelor of Science in Computer Science

Wilfrid Laurier University

April 2028

Waterloo, ON

- **Core Coursework:** Data Structures, Algorithms, OOP, Discrete Mathematics, Computer Architecture, Digital Electronics

Technical Experience

Full-Stack Developer — Team Lead

ConHacks | InstantService

April 2026

Waterloo, ON

- **Led a team of 4 developers to architect an AI-native dispatch engine** for InstantService, reducing client service-acquisition time by **70%** via automated request classification using **Gemini 1.5 Pro**.
- **Engineered a multi-weighted contractor matching algorithm** with FastAPI and Snowflake, optimizing contractor allocation to drive a **40% increase in job volume** and deliver **instant, under-3-min service confirmations**.
- **Architected a secure backend infrastructure** for tier-based pricing and contractor verification, automating verification of **license/insurance certificates** to establish a high-trust production environment.

Operations Logistics

Hack the North (Canada's Largest Hackathon)

Sept 2021 – Sept 2022

Waterloo, ON

- **Facilitated a seamless 36-hour event experience** for **1,000+** global students, managing on-site logistics, equipment layouts, and wayfinding operations.
- **Orchestrated attendee registration and technical demo scheduling** for **827** attendees across **222** concurrent projects, resolving critical operational bottlenecks with zero major schedule disruptions.
- **Streamlined equipment layout and resolved operational bottlenecks** rapidly, ensuring zero major schedule disruptions for operations backed by major corporate sponsors.

High-Impact Projects

InstantService | AI-Powered Dispatch PWA | [GitHub](#)

Next.js | FastAPI | Snowflake | Gemini | ElevenLabs

- **Developed a mobile-first PWA** with a real-time feedback loop, integrating **ElevenLabs Neural Speech** to deliver immediate AI-generated verbal confirmations synced to a **Snowflake** data layer.
- **Integrated Google Maps APIs** and live **WebSocket** feeds to calculate optimal service paths, reducing overall API overhead by **20%**.
- **Implemented a multi-tier service architecture** that dynamically matches clients with pros based on trust scores, reducing friction in service acquisition for emergency repair requests.

Subscription Manager | Full-Stack MVC API | [GitHub](#)

Node.js | Express | MongoDB | Arcjet | Upstash

- **Built a production-ready MVC API** using **Node.js/Express**, implementing **JWT-based authentication** and **Zod** schemas to validate inputs and eliminate 95% of common payload errors.
- **Hardened API security** by deploying to **Render** with **Arcjet** rate limiting and bot detection, maintaining 99.9% uptime against malicious traffic.
- **Automated lifecycle notifications** using **Upstash Workflow** and **Nodemailer**, ensuring 100% reliability for event-driven trial expirations.

MSA Website | Digital Engagement Platform | [Live Site](#)

Next.js | TypeScript | Payload CMS

- **Improved navigation flows and content accessibility** for a **250+** member community platform, consulting on UI/UX product design for the official **React/Next.js** digital ecosystem.
- **Enhanced core community features** by optimizing validation logic and content structure for live prayer timings, halal directories, and interactive event registration forms.
- **Developed interactive event registration and fundraising modules**, supporting community-driven initiatives that generated **\$50K+** in donations.

Technical Skills Summary

Languages: Python, Java, C, JavaScript, TypeScript, HTML, CSS, SQL

Frameworks & UI: React.js, Next.js, PWA, Node.js, Tailwind CSS, Docker, shadcn/ui, Lucide Icons

AI & Backend: FastAPI, Gemini 1.5 Pro, ElevenLabs, Arcjet Security, Upstash Workflow, RESTful APIs

Cloud & Data: Snowflake, MongoDB, DigitalOcean, Vercel, Render, Git, GitHub, MVC Architecture

Concepts: System Architecture, AI-Native Development, Latency Optimization, Data Warehousing, CI/CD